

BeliefLens: Belief Dynamics in Real LLM Forecasting Systems

Four production harnesses, one date-gated sandbox

BeliefLens Project

Deep Digital Intelligence · <https://github.com/DDI-droid/beliefLens>

September 11, 2026 · Experiment 1, v2

Abstract

LLM forecasters are *judgmental* forecasters: behind every stated probability is something that behaves like a belief, and behind every revision an update rule. This report measures both, for four **real forecasting systems** rather than prompt-level imitations of them. We take the orchestration code of Analytica, of the FutureSim benchmark’s reference agent, and of the BLF sequential-Bayesian forecaster, trim each only where the sandbox requires, and run them alongside ReAct—which needs no adaptation, being a prompt pattern over a tool loop—inside a date-gated replay environment built on 339,801 news articles. Six binary questions, drawn as three matched pairs from the FutureSim benchmark’s own question set, are forecast under two protocols: **sequential** (a trajectory of dates, the system’s own state carried forward, $k=5$ rollouts) and **parallel** (a fresh one-shot run per date with no history, $k=5$ rollouts), so that every belief is a distribution and the effect of carrying history is isolated by construction. A sixth date, placed two days *after* each question resolves, serves as an **outcome probe**: by then the corpus contains the answer, so it measures whether a system moves to the truth when the truth is available. Findings. Update gain—how far a carried belief moves when the evidence moves—separates the four systems by a factor of nine, from 0.08 to 0.74. Analytica’s architecture makes it the most rigid tracker and simultaneously the most internally coherent, producing near-additive paired beliefs without ever seeing both halves of a pair. The outcome probe reveals an inversion invisible to any trajectory metric: the system that tracks evidence best under uncertainty (**blf-full**, gain 0.74) is the worst at absorbing a decided outcome, landing 0.46 from the truth with the answer in its own search results, while the system that searches least (**futureSim-full**, 1.3 searches per forecast) resolves five of six questions exactly. In both cases the behaviour is traceable to a specific line of architecture rather than to model capability. All 1,432 logged searches in the audit replay with zero articles past the date gate.

1 Introduction

Benchmarks score LLM forecasting agents on outcomes, and observe behaviours—anchoring to initial forecasts, “self-conditioning” on past rationales—that are claims about *belief dynamics*. But a score cannot say where a pathology lives: in what the agent *believes*, in how it *updates*, or in how it *gathers evidence*. Those are three different failures with three different fixes, and one number per condition cannot separate them.

This report separates them, for systems as they are actually built. That last clause is the methodological commitment. It is easy, and common, to study “a ReAct-style agent” or “an Analytica-style decomposition” by writing a prompt that describes the reasoning shape and running it on a shared loop. We did exactly that in an earlier iteration of this project, and it produced results that did not survive contact with the real systems: when a structure is *requested in a prompt* the model may quietly decline it, and when the same structure is *enforced in orchestration code* it cannot. Everything reported here runs the orchestration.

Contributions.

- **Four real systems in one sandbox.** Analytica’s analyzer/grounder/synthesizer pipeline with its linear composition computed in code; the FutureSim benchmark’s own reference agent with its session

architecture, pandas-backed task state and CRUD memory; the BLF forecaster’s Algorithm 1 with K -trial shrunken logit-mean aggregation; and ReAct. Each is trimmed only where the sandbox requires, and every trim is stated (§4).

- **A protocol that isolates history.** Sequential and parallel arms over an identical date grid, $k=5$ rollouts each, so the effect of carrying state is a measured difference rather than an inference (§5).
- **The outcome probe.** A date placed after resolution, when the corpus contains the answer. It converts “is this system rigid?” from a relative question into an absolute one, and it exposes a tracking/resolving inversion that no trajectory metric can see (§7).
- **Matched question pairs.** Three two-sided contests binarised into both finalists, so coherence— $P(a) + P(b)$, which must not exceed 1 and should approach it as the field narrows—is measurable on every group, and the resolved answers are balanced three YES to three NO (§3.3).

2 Background: the source systems

Analytica [1]. Three agent roles, two algorithms. ANALYZE expands a root proposition into a tree of sub-propositions until a leaf budget. SYNTHESIZE walks the tree bottom-up: *Grounder* agents evaluate all leaves *in parallel*, each returning a soft truth value and a written report, and a *Synthesizer* scores each non-leaf from its children under a **linear** rule $p = \beta_0 + \sum_j \beta_j p_j$, where the LLM emits the coefficients and the arithmetic is not the model’s job. Their §4.1 also defines **resynthesis**: after grounding, a user may edit any node and the system recomputes only the affected branches. Our continual-forecasting extension is exactly that mechanism driven by a calendar.

FutureSim [2]. A chronological, replayable simulation. The environment enforces two actions, `submit_forecast` and `next_day`; everything else is the agent’s business. Their reference agent, `basicAgent`, takes one tool call per turn, chosen from `query_df`, `search_news`, four memory CRUD tools, `submit_forecasts` and `next_day`. It inspects the question table by writing Python in a sandbox, and has its context cleared between sessions so that structured memory and past predictions are all that survive. Two of their findings motivate this work: agents started from a poor initial forecast set fail to adapt away from it, and the same agent asked each question *directly* one day before resolution outperforms itself walking the same information forward day by day.

BLF [5]. An agentic system for *binary* question forecasting. Its belief is a JSON object—probability, confidence, evidence for and against, open uncertainties—and each step of its loop returns *both* the next action and the updated belief in one generation. $T_{\max}=10$; submitted probabilities clamp to $[0.05, 0.95]$. Around the loop sit $K=5$ independent trials aggregated by a shrunken logit-space mean $\hat{p} = \sigma(\alpha \cdot \frac{1}{K} \sum_k \text{logit}(p_k))$ with α falling as between-trial variance rises, plus hierarchical Platt calibration.

ReAct [4] interleaves thinking and acting with no imposed structure. Unlike the other three it *is* a prompt pattern over a tool loop, so there is nothing to re-implement: the harness here is its source system, exposed one date at a time.

Agent-BRACE [3] supplies the framing question—is an externalised belief structure the agent’s *working* belief, or a report it writes alongside one?—but is a training method, not an installable harness, so it is not run.

3 Infrastructure

3.1 Corpus and date gate

Articles are the FutureSim corpus (HuggingFace `shash42/forecast-news`; CCNews-derived, deduplicated, dated), sliced to the simulation window: 2025-12-01–2026-03-31, indexed to SQLite FTS5 (BM25), 339,801

rows. At simulation date D the search tool returns only articles with date $\leq D$; **the constraint is in the SQL, not in the prompt**. The audit re-executes every logged search against its own turn’s date; zero returned articles postdate the gate.

We do not use FutureSim’s prebuilt hybrid index. Retrieval here is BM25, weaker than theirs, but byte-identical across all four harnesses—a constant of the experiment rather than a confound in it.

3.2 Model choice

All four harnesses run on `gpt-5-mini` (released 2025-08-07). This is a validity decision, not a cost decision: the questions resolve January–February 2026, so a later model would *recall* rather than forecast, and a model that already knows the outcome does not move its belief—destroying the thing being measured.

3.3 Questions

Why binary. Every measurement here needs the belief to be a single number in $[0, 1]$: update gain is a regression slope of Δp on Δp , the anchor gap is a difference of two probabilities, placement is a percentile within a set of draws. FutureSim’s native format is a distribution over outcomes the agent invents for itself, whose support changes between dates; no scalar trajectory exists in that world. Two of the three source systems are binary-native—BLF states it outright, Agent-BRACE attaches a probability to a claim—so the choice is faithful to them and unfaithful to FutureSim’s task format, and we say so. The consequence is that our numbers are not comparable to FutureSim’s benchmark scores.

The groups. Each group is one free-form question from the FutureSim benchmark split (`aljazeera2026Q1`, 330 questions) converted into *two* binary questions, one per finalist of a two-sided contest. The runner-up in each is read off the source article, not supplied from memory.

group	question id	binary question, verbatim as the harness sees it	resolves
hockey	hockey_usa	Will the United States win the men’s ice hockey gold medal at the Milano Cortina 2026 Winter Olympics?	YES
	hockey_can	Will Canada win the men’s ice hockey gold medal at the Milano Cortina 2026 Winter Olympics?	NO
ausopen	ausopen_alcaraz	Will Carlos Alcaraz win the 2026 Australian Open men’s singles title?	YES
	ausopen_djokovic	Will Novak Djokovic win the 2026 Australian Open men’s singles title?	NO
superbowl	superbowl_sea	Will the Seattle Seahawks win Super Bowl LX?	YES
	superbowl_ne	Will the New England Patriots win Super Bowl LX?	NO

Table 1: The six questions, three resolving YES and three NO. Each group binarises one free-form question from the benchmark split—“*Which country will win the men’s ice hockey gold medal...?*”, “*Who will win the 2026 Australian Open men’s singles title?*”, “*Which NFL team will win Super Bowl LX?*”—into its two finalists. Each question is put to the harness on its own, fully self-contained, with background text that names the event and date but neither finalist, so a harness forecasting one half of a pair has no way to know the other half exists. That is what makes coherence an observation about the system rather than an artefact of the setup.

The date grid. Six dates per question, as offsets from the resolution date R . The forecasting trajectory is D_1 at $R-39$, then $R-25$, $R-14$, $R-7$ and D_5 at $R-2$, accelerating toward the event; then D_6 at $R+2$, the **outcome probe**.

4 The four harnesses

Each system runs its own orchestration. The LLM is confined to the roles its paper gives it, and the composition, control flow and state are executed in code. What follows is what each system is, and precisely

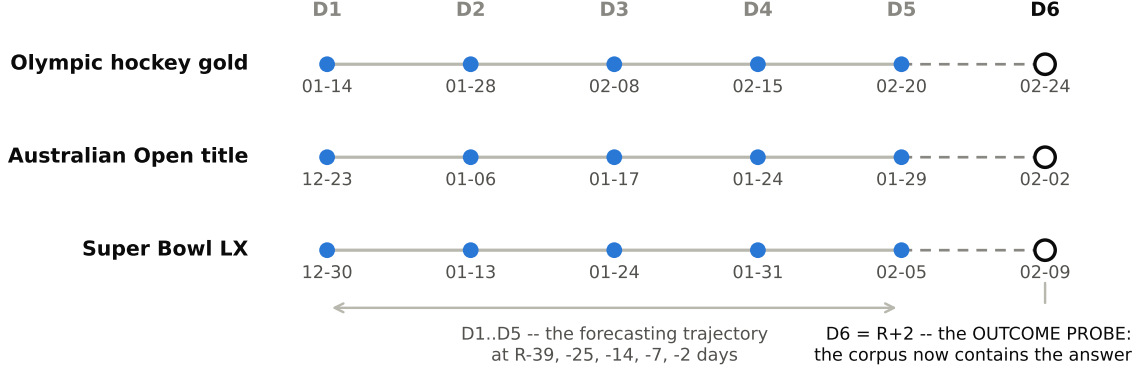


Figure 1: The date grid. Five trajectory dates per question, then a probe date two days after resolution. D_6 is a diagnostic, not part of the trajectory: by then the corpus contains the result, so every dynamics metric in this report is computed on $D_1..D_5$ only. Mixing a post-outcome date into a belief-dynamics slope would corrupt it.

what we changed.

4.1 analytica-full

An *analyzer* agent decomposes the question into soft sub-propositions (leaf budget 8, up to three expansion rounds on the first date); parallel per-leaf *grounder* agents each generate search queries, read gated articles and conclude a leaf probability with a written report in a separate second call; a *synthesizer* agent assigns weights; and the composition $p = \text{clamp}(\beta_0 + \sum_i \beta_i p_i)$ is computed recursively in *Python*. The model never states the final probability.

The paper is single-shot, so continual forecasting needs an extension—and the extension is the paper’s own resynthesis mechanism driven by a calendar. The tree is carried as state; each date runs an analyzer **edit pass**; every leaf is re-grounded but **delta-grounded**, its search window restricted to the period since *that leaf* last looked and its own previous value shown to it; synthesizer weights are re-elicited **only** where a node’s children changed.

4.2 blf-full

Algorithm 1 verbatim in structure: an explicit JSON belief state; each step one generation returning both an action and the updated belief; $T_{\max}=10$; the submitted probability clamped to $[0.05, 0.95]$; $K=5$ trials per date aggregated by the paper’s shrunken logit-space mean with $\alpha = 1/(1 + \text{Var}(\text{logit}(p_k)))$ —our instantiation of their variance-dependent α , whose functional form they do not pin down. The belief state of the median-probability trial carries across dates in SEQ and is rebuilt from scratch in PAR. Their **LeakFilter**, a second-pass LLM filter on search results, is replaced by our SQL date gate, which is strictly stronger. Platt calibration is omitted: it requires resolved training questions, and the window contains none.

4.3 futuresim-full

The FutureSim benchmark’s own reference agent. Context is cleared between dates, so structured memory and past predictions are all that survive; one tool call per turn from their action set; **query_df** is a real pandas sandbox over the question table with **pd** in scope; memory is four CRUD tools plus a separate end-of-session memory-revision phase; **submit_forecasts** takes a distribution over outcomes, from which we read $P(\text{Yes})$.

Two changes. We drive the calendar rather than letting the agent advance it, so that all four harnesses share one date grid and remain comparable. And the agent must submit on every date: their system permits

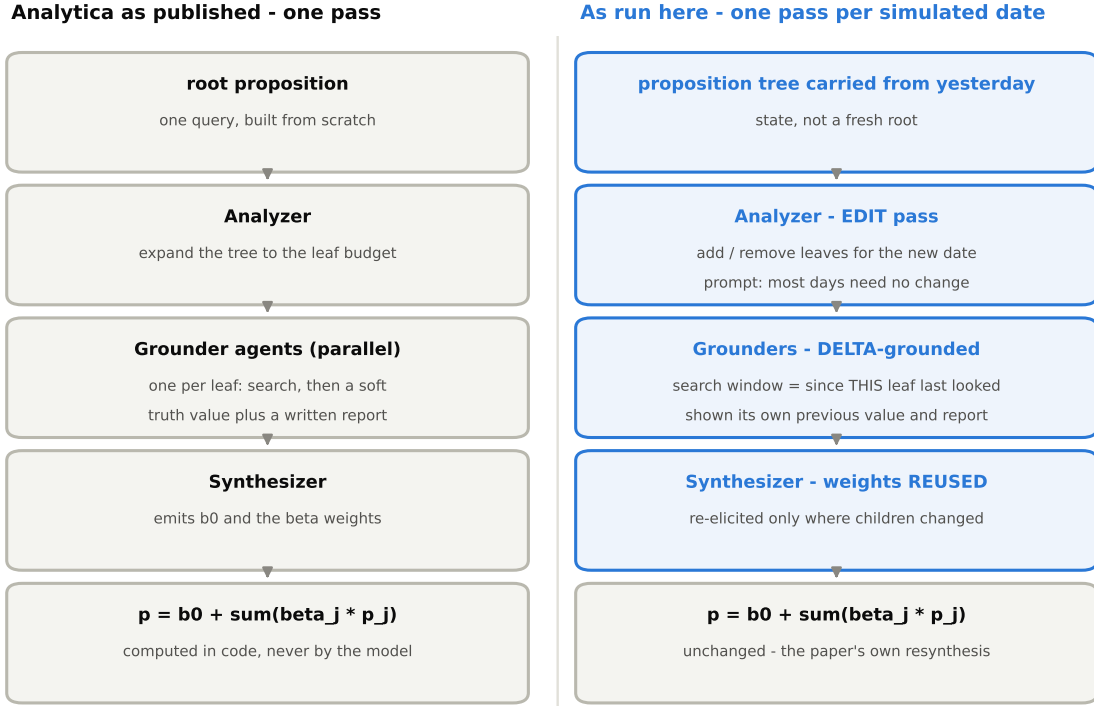


Figure 2: Analytica, published versus as run. Highlighted boxes are the whole continual-forecasting delta; the composition arithmetic is unchanged.

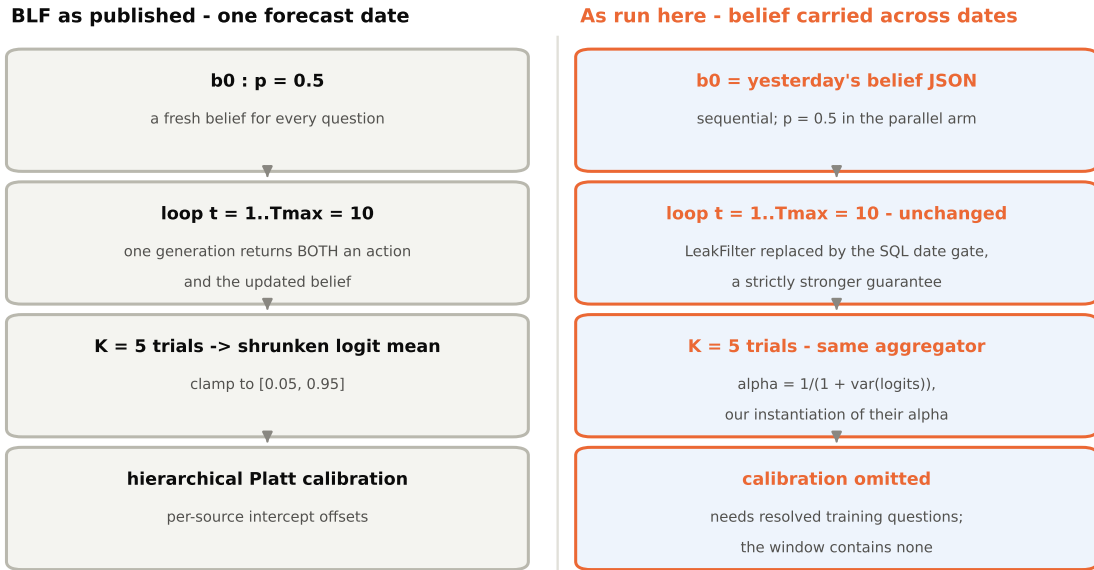


Figure 3: BLF, published versus as run. Note that K is *internal*: the five trials collapse to one number, which is that date's forecast. It is not our rollout count.

ending a session without submitting, leaving the standing forecast in place, and disabling that is the single largest liberty this report takes with a source system.

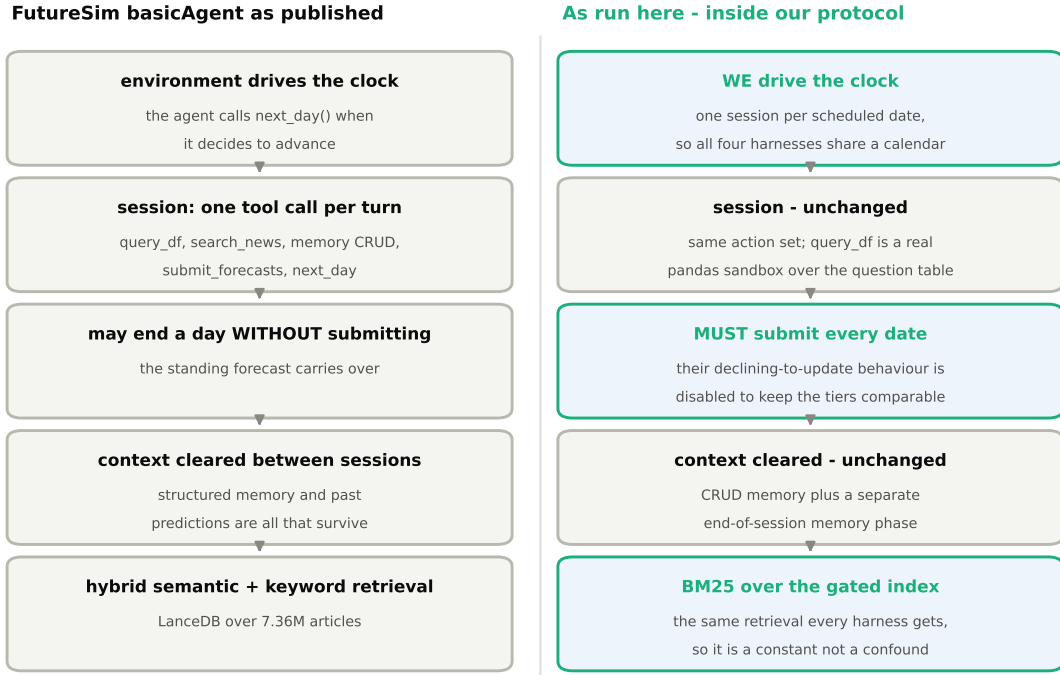


Figure 4: FutureSim’s reference agent, published versus as run.

4.4 react

A think/search/read loop with no imposed structure, at most six search rounds, then a forced answer. ReAct is a prompt pattern over a tool loop, so this harness *is* its source system; the only adaptation is exposing it one date at a time.

5 Protocols and metrics

Sequential (seq): the five trajectory dates in order; each date the harness sees its own previous forecasts and whatever state its architecture carries (Analytica’s tree, BLF’s belief JSON, FutureSim’s memory); $k=5$ independent rollouts of the whole trajectory. **Parallel (par):** for each date, a fresh run given all information up to that date and *no* history; $k=5$ rollouts per date. The parallel arm is what the evidence alone supports; the difference between the arms is what carrying history does.

anchor gap

$SEQ_t - \overline{PAR}_t$, signed, for $t \geq 2$. What carrying history does relative to evidence alone.

lag advantage

$|SEQ_t - \overline{PAR}_t| - |SEQ_t - \overline{PAR}_{t-1}|$. Positive means the sequence tracks *yesterday’s* evidence-conditioned belief better than today’s.

update gain

OLS slope of ΔSEQ_t on $\Delta \overline{PAR}_t$. 1 means the carried belief moves exactly as far as the evidence moves; < 1 is conservatism; < 0 moves against it.

placement (PIT)

Percentile of each sequential forecast within that date’s parallel draws; 50 is unbiased.

stabilisation

$\text{sd}(\text{SEQ})/\text{sd}(\text{PAR})$ per date. Below 1 means carrying history reduces variance across rollouts.

coherence

$P(a) + P(b)$ within a matched group. Must not exceed 1; should approach it as the field narrows to the two candidates.

total movement

$|\Delta|_{\text{tot}}$, the summed absolute day-to-day change of a rollout.

Brier The plain quadratic score $\frac{1}{n} \sum_i (p_i - y_i)^2$ with $y_i \in \{0, 1\}$ —not a skill score. 0 is perfect and **0.25** is the constant- $\frac{1}{2}$ forecaster.

Every metric above is computed on $D_1..D_5$. The probe date is reported separately in §7.

6 Results

harness	$ \Delta _{\text{tot}}$	gap	$ \text{gap} $	gain	PIT_{med}	lag adv.	searches SEQ/PAR
analytica-full	0.243	−0.006	0.128	0.08	40	+0.016	6.8 / 6.6
blf-full	0.184	−0.031	0.047	0.74	0	−0.012	22.8 / 25.2
futuresim-full	0.274	+0.029	0.079	0.45	60	−0.023	1.3 / 1.2
react	0.240	−0.012	0.067	0.45	40	−0.019	2.6 / 2.6

Table 2: Headline metrics, $k=5$ in both arms, pooled over six questions and five trajectory dates ($n=1,200$ forecasts). Bold marks the extreme of each column.

6.1 Update gain separates the systems by a factor of nine

The single most discriminating number in the table is update gain, and it ranges from 0.08 to 0.74. **analytica-full** moves its carried belief eight points for every hundred points the evidence moves. **blf-full** moves seventy-four.

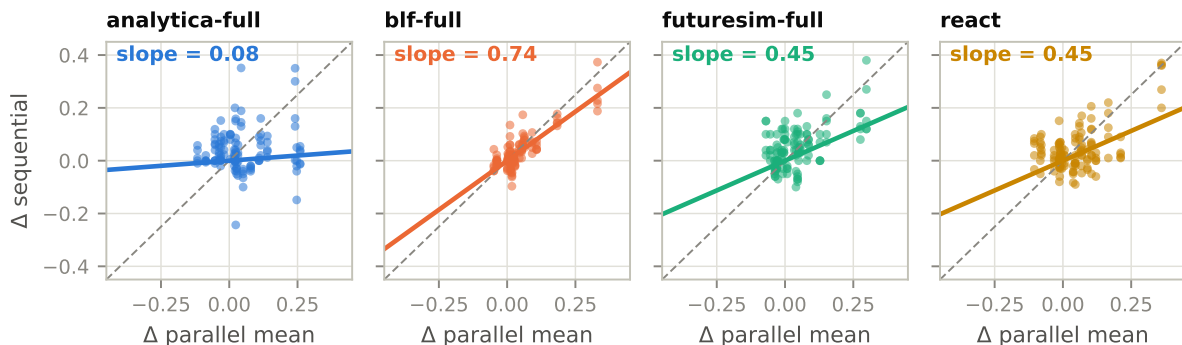


Figure 5: Update gain. Each dot is one date-step: x is the movement of the fresh-eyes mean, y the movement of a sequential rollout; the dashed diagonal is $y=x$, perfect tracking.

Analytica’s rigidity is traceable to three specific lines of its architecture, none of which is an instruction a model may decline. A leaf that searches only news *since it last looked* cannot react to standing evidence it has already discounted. A leaf handed its own prior value is anchored by design. And a parent whose children did not change reuses yesterday’s weights outright. The belief is structurally prevented from responding to evidence that does not survive the edit pass.

A result that did not replicate. An earlier iteration of this experiment, run on two questions with $K=3$, measured **blf-full**’s update gain at -0.17 and reported it as evidence that the system moved *against* the evidence—flagged at the time as weakly identified, because its fresh-eyes mean rarely moved more than ± 0.06 and the slope was fitted over a very narrow range. At six questions and $K=5$ the same measurement gives $+0.74$. The earlier number was not merely imprecise, it had the wrong sign. We report this prominently because it is the clearest available evidence that two questions cannot support a claim of this kind, and because the failure mode—a slope confidently estimated over a degenerate x -range—is not specific to this project.

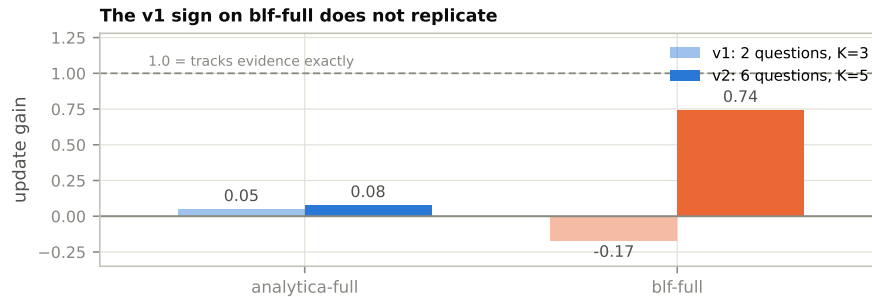


Figure 6: Non-replication. The same metric, the same harness, the same code path; two questions at $K=3$ against six questions at $K=5$.

6.2 Anchoring, placement and stabilisation

The signed anchor gaps are small for every system ($|\cdot| \leq 0.031$), so none of them holds a systematically higher or lower belief than fresh eyes would. What differs is dispersion: **analytica-full**’s *absolute* gap is 0.128, nearly three times **blf-full**’s 0.047, so its carried belief wanders far from the evidence-conditioned one in both directions without a net bias.

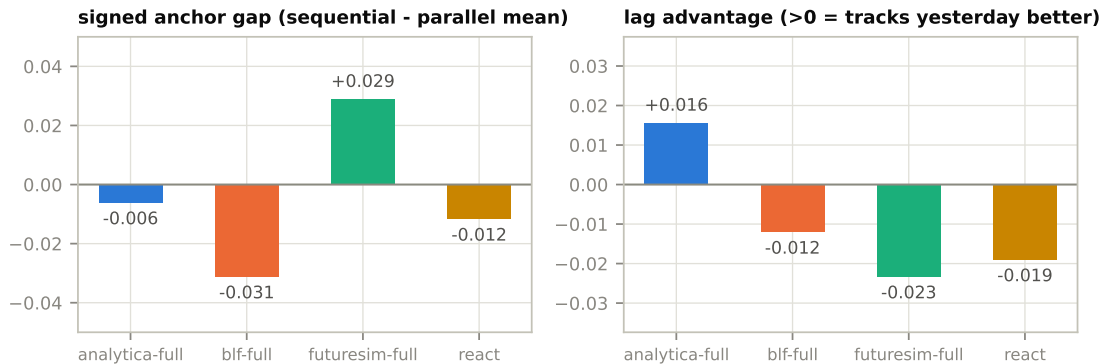


Figure 7: Anchoring summary. Left: signed anchor gap. Right: lag advantage; only **analytica-full** is positive, and only slightly.

blf-full’s placement median of 0 deserves care. It means its sequential forecast sits at or below *every* parallel draw, essentially always. That sounds extreme until one notices its $K=5$ aggregation makes the parallel draws exceptionally tight—so a signed gap of only -0.031 is enough to fall below all five. The placement statistic is measuring the aggregator’s variance reduction as much as the belief’s bias, and should not be read as strong anchoring.

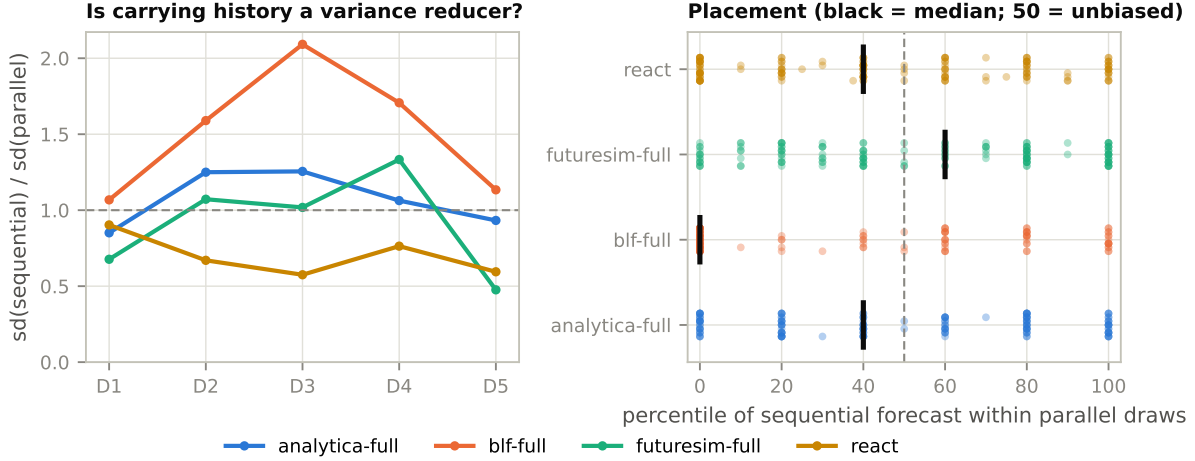


Figure 8: Left: stabilisation ratio. **react** is the only system consistently below 1: carrying history genuinely reduces its cross-rollout variance. **blf-full** sits above 1 throughout, peaking at 2.09—carrying the belief state makes its rollouts *diverge*, consistent with the carried JSON acting as a strong per-rollout prior that each trajectory then drags along its own path. **Right:** placement.

6.3 Coherence comes from architecture

analytica-full produces near-additive paired beliefs on the hockey group—1.00, 0.98, 1.01, 1.02, 0.99 across the five dates—*without ever seeing the two questions together*. Decomposition into sub-propositions plus linear synthesis yields consistency as a side effect of the architecture. No other system comes close on that group; **react** sits at 0.67 and **blf-full** at 0.74.

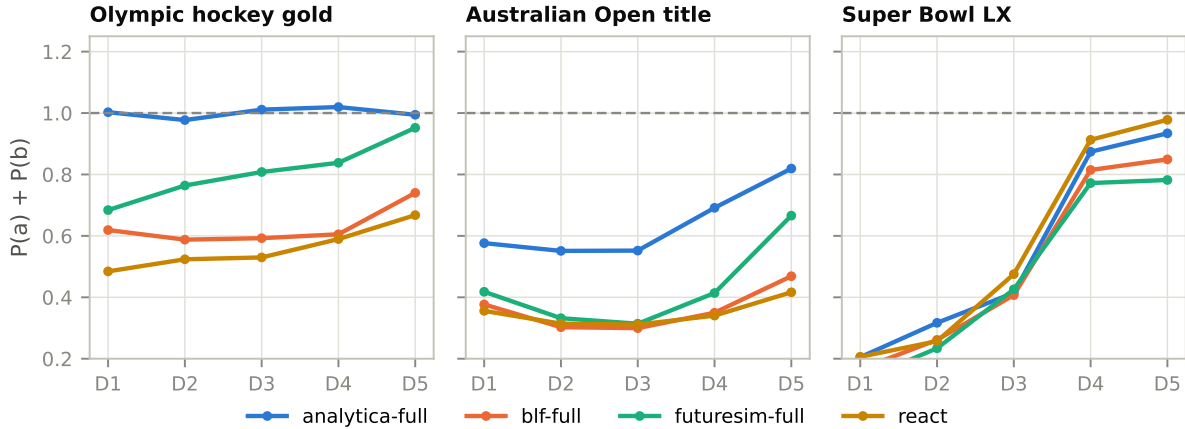


Figure 9: Coherence per group, $P(a) + P(b)$ against the target of 1. The Super Bowl group shows every system converging late, as the playoff field narrows to two teams—the sum climbs from about 0.2 at D_1 to 0.78–0.98 at D_5 . That is the metric behaving correctly: before the finalists are known the sum *should* be well below 1.

6.4 Accuracy

All four systems improve monotonically across the window, from a pooled Brier around 0.38–0.40 at D_1 to 0.24–0.32 at D_5 . **blf-full** finishes at **0.240**, below the constant- $\frac{1}{2}$ baseline of 0.25; the others do not clear it.

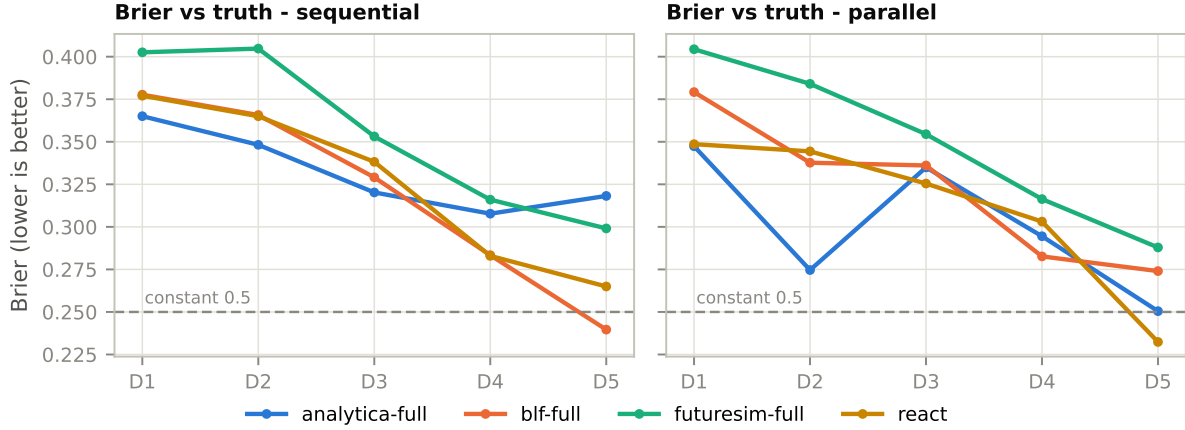


Figure 10: Brier against the resolved outcomes, pooled over the six questions.

6.5 Search effort varies by a factor of eighteen

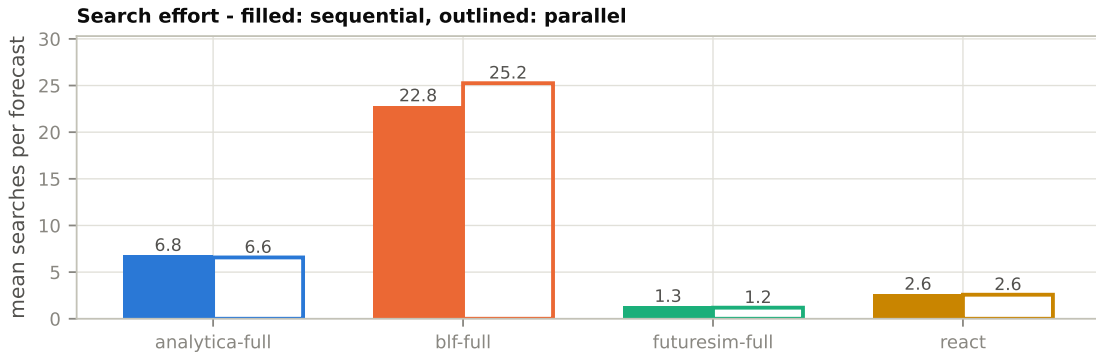


Figure 11: Search effort. blf-full issues 22.8 searches per forecast—five trials times roughly five queries each—while futuresim-full issues 1.3.

This is where the systems differ most starkly, and the naive reading—that more search means better forecasting—is wrong in both directions, as the next section shows.

7 The outcome probe

At $D_6 = R+2$ the corpus contains the result. A system that cannot reach the truth here is not failing at forecasting; it is failing at reading. This turns rigidity from a relative measurement into an absolute one.

harness	mean move toward truth	mean forecast – truth	direction correct
futuresim-full	+0.498	0.025	6/6
react	+0.329	0.134	6/6
analytica-full	+0.252	0.280	6/6
blf-full	+0.014	0.462	3/6

Table 3: Outcome probe. futuresim-full lands on the truth *exactly*—0.000 or 1.000—on five of the six questions, and at 0.85 on the sixth. blf-full barely moves at all.

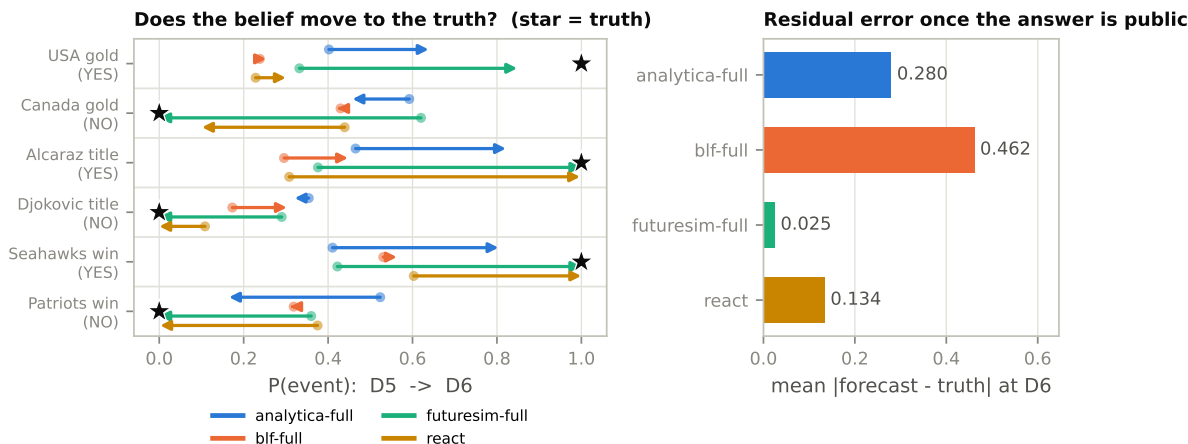


Figure 12: Left: each arrow runs from a system’s mean belief at D_5 to its mean belief at D_6 ; the star is the truth. **Right:** residual error once the answer is public.

The inversion. `blf-full` has the best update gain of any system measured here (0.74) and the worst outcome-probe error (0.462, correct direction on only three of six questions). It is simultaneously the best tracker of uncertain evidence and the worst absorber of decided fact. The mechanism is not mysterious: its forecast is a shrunken logit-space mean over five trials, clamped to $[0.05, 0.95]$. The clamp alone caps its error at 0.05 from either extreme, and the variance shrinkage pulls hardest toward the prior exactly when trials disagree. The machinery that makes it responsive under uncertainty is the same machinery that prevents it from committing under certainty.

The other direction. `futuresim-full` issues 1.3 searches per forecast, by far the fewest, and resolves five of six questions exactly. Its low search count is not laziness. When the signal is decisive one search suffices; the cost is paid earlier in the window, where it has the weakest Brier of the four at D_1 – D_2 . Search volume is a poor proxy for evidence use in either direction.

8 Synthesis

Three claims survive the six questions.

Rigidity is architectural, and locatable. The nine-fold spread in update gain is not a spread in model capability—every system runs the same model on the same corpus behind the same date gate. Analytica’s delta-grounding, weight reuse and per-leaf anchoring produce a gain of 0.08; BLF’s flat belief with full re-search produces 0.74. Pointing at the mechanism is possible because the mechanism is code.

Tracking and resolving are different capabilities. No trajectory metric distinguishes a system that is appropriately uncertain from one that is structurally incapable of confidence. The probe does, and it ranks the systems almost in reverse.

Consistency can be a side effect of structure. Analytica’s paired beliefs sum to approximately 1 on the hockey group with no mechanism that could have told it a pair existed. If coherence is wanted, it appears to be obtainable by construction rather than by instruction.

9 Reproducibility and the data package

Everything below ships in `results/exp1_v2/`.

Traces are per-harness and preserve each system’s internal state, which is what makes this dataset reusable rather than merely a table of probabilities:

file	contents	size
<code>e1_runs.jsonl</code>	120 sequential chains $\times$ 6 dates, with full traces	5.2 MB
<code>e1_independent.jsonl</code>	720 parallel singles, with full traces	5.4 MB
<code>metrics_v2.json</code>	every metric in this report	85 KB
<code>usage.json</code>	call and token ledger	—

Table 4: Collected data. 713 of 720 sequential turns and 718 of 720 parallel singles carry a complete trace; the gaps are exactly the rate-limited forecasts of §10.

- **analytica-full:** the full proposition tree at every date—each node’s statement, its grounder’s written report, its soft truth value, the synthesizer’s β weights and intercept, and the date it was last grounded—plus the edit list and every search issued.
- **blf-full:** the per-trial probabilities, the aggregate, the belief JSON (probability, confidence, evidence for and against, open uncertainties), and every search tagged by trial and step.
- **futuresim-full:** the action sequence, the `query_df` code the agent wrote and what it returned, the memory store and every operation applied to it, and its searches.
- **react:** the reasoning text, the searches, and whether the answer was forced by exhausting the search budget.

Reproduce with `run_real_e1.py` for the run, `analyze_v2.py` for the metrics and `figures_v2.py` for the figures; `watch_run.py` monitors a run in progress. The key is read from `.env`.

Cost. 19,385 model calls, 35.1M input and 8.2M output tokens recorded, 2h22m at twelve workers, against a hard 32,000-call cap enforced in the call wrapper. The recorded token figures exclude **react**, which calls through its own client; its calls are counted but its tokens are not, so the true token total is roughly 10–15% higher.

10 Limitations

1. **Six questions, three groups.** Better than two, and enough to overturn a two-question result (§6), but not enough to support fine distinctions. The gain ordering $0.08 < 0.45 \approx 0.45 < 0.74$ is coarse and we do not read the tie between **futuresim-full** and **react** as meaningful.
2. **All three contests are sport.** They share a structure—a narrowing field, a decisive public result—that political or economic questions do not. The coherence convergence in particular may be specific to bracket-shaped events.
3. $k=5$ **rollouts** bounds every distributional claim; placement has five-draw granularity, which is why a median of exactly 0 is possible.
4. **Draws are ordinary decoding variation.** Current reasoning endpoints reject temperature control.
5. **Stated beliefs only.** Everything here is a consistency property of verbalised probabilities under intervention, not a claim about internal representations.
6. **Retrieval is BM25**, weaker than FutureSim’s hybrid index. Held constant across harnesses, so comparisons hold, but absolute accuracy is a lower bound for all four.
7. **futuresim-full is forced to submit every date.** Its source system may decline to update, and that behaviour—which FutureSim’s own anchoring finding is partly about—is disabled here to keep the harnesses comparable. It is the largest liberty this report takes.
8. **Nine forecasts of 1,440 (0.6%) were lost** to rate-limit errors that outlived five retries; 1,780 retries were absorbed successfully. The losses are scattered and no cell lost more than one rollout.

9. **Binarisation changes the task.** Our numbers are not comparable to FutureSim benchmark scores, and binary questions are easier than the open-ended originals.

11 Conclusion

Running four forecasting systems as they are actually built, rather than as prompts describing them, makes their belief dynamics attributable to specific architecture: delta-grounding and weight reuse produce rigidity, a flat belief with full re-search produces responsiveness, and shrunk multi-trial aggregation produces an inability to commit when the evidence is decisive. A date placed after resolution turns the last of these from an inference into a measurement, and reveals that the best tracker of uncertain evidence is the worst absorber of certain evidence. The collected traces preserve each system’s internal state at every date, so the same run supports analysis well beyond the metrics reported here.

References

- [1] *Analytica: Soft Propositional Reasoning for Robust and Scalable LLM-Driven Analysis*. arXiv:2604.23072, 2026.
- [2] *FutureSim: Replaying World Events to Evaluate Adaptive Agents*. arXiv:2605.15188, 2026.
- [3] *Agent-BRACE: Decoupling Beliefs from Actions in Long-Horizon Tasks via Verbalized State Uncertainty*. arXiv:2605.11436, 2026.
- [4] *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR, 2023.
- [5] *Agentic Forecasting using Sequential Bayesian Updating of Linguistic Beliefs*. arXiv:2604.18576, 2026.
- [6] *Verification of forecasts expressed in terms of probability*. Monthly Weather Review, 1950.

A Belief trajectories, all four systems

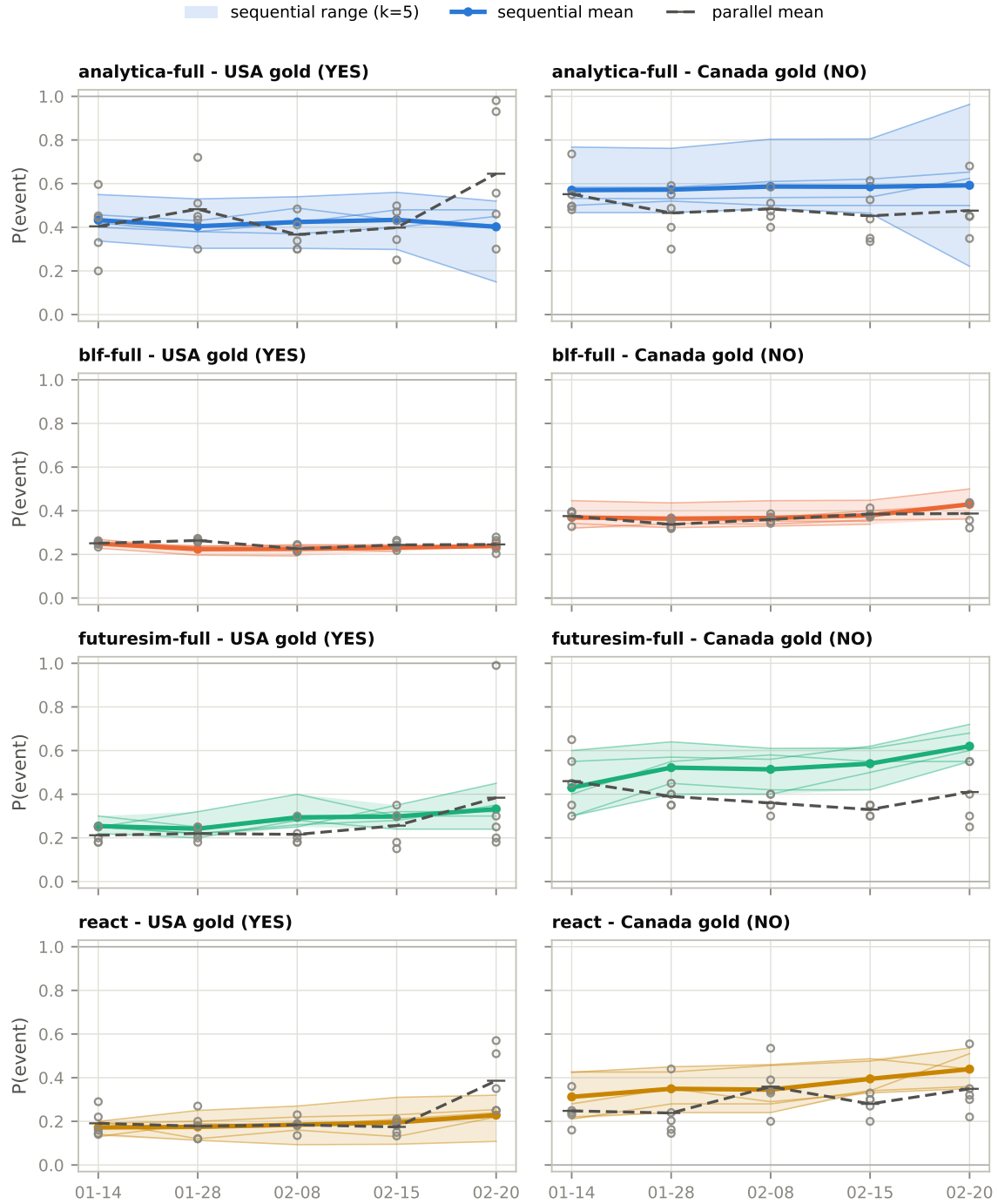


Figure 13: Olympic hockey gold. Shaded band and bold line: range and mean of the five sequential rollouts. Open dots and dashed line: the five parallel draws per date and their mean. Horizontal line: the resolved truth.

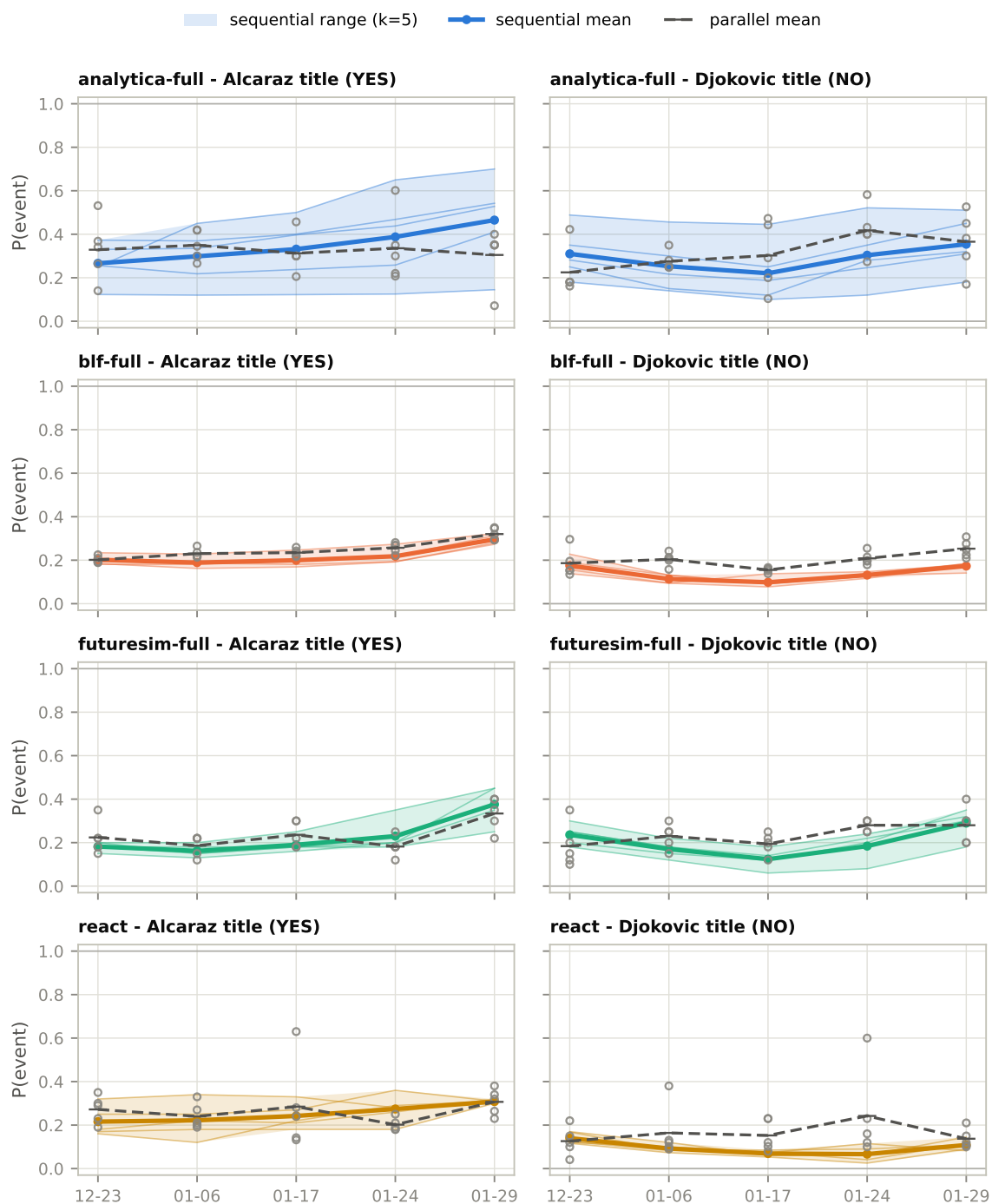


Figure 14: Australian Open title.

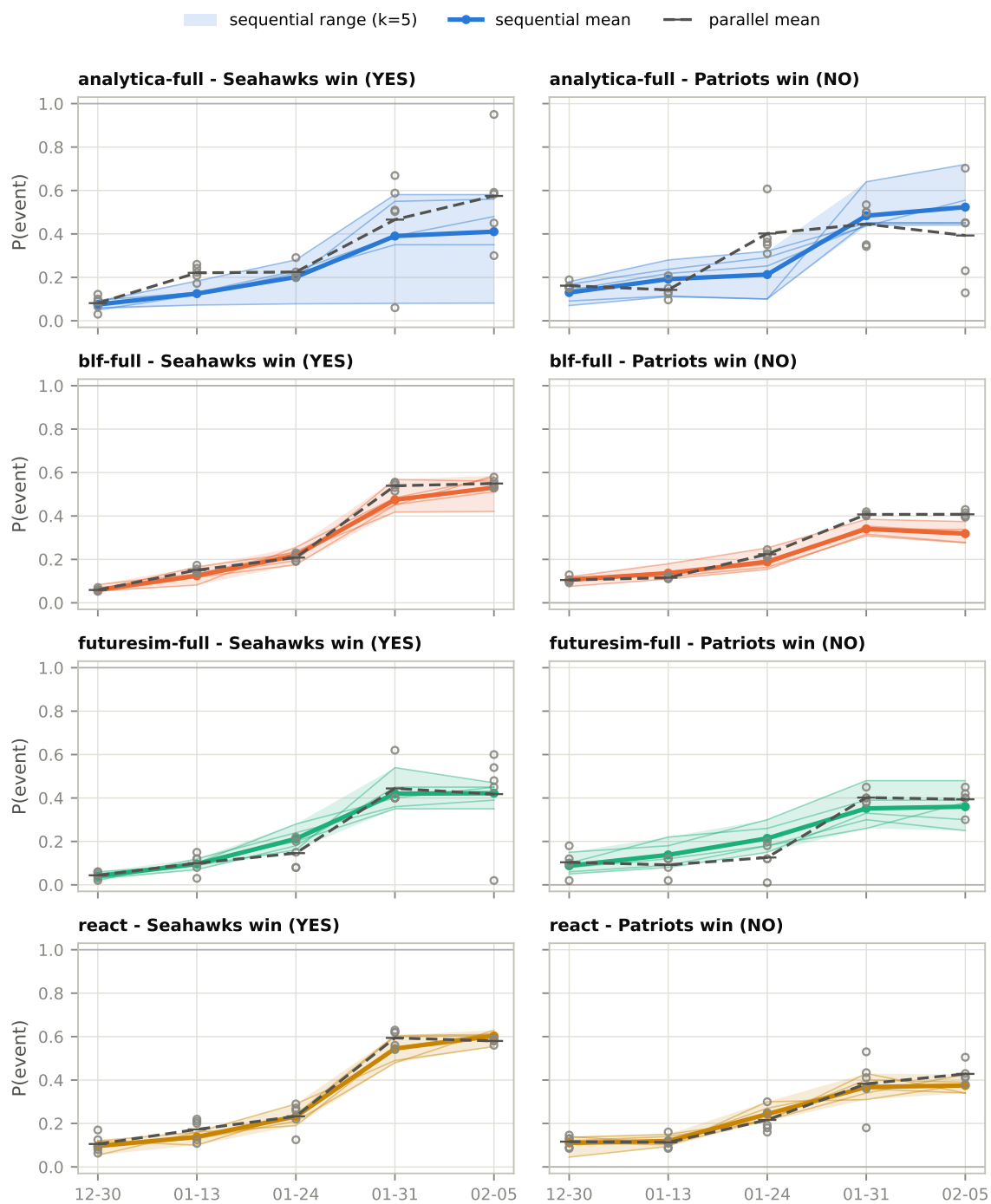


Figure 15: Super Bowl LX.

B Per-question outcome probe

harness	question	truth	D_5	D_6
analytica-full	hockey_usa	1	0.40	0.64
	hockey_can	0	0.59	0.46
	ausopen_alcaraz	1	0.47	0.82
	ausopen_djokovic	0	0.35	0.32
	superbowl_sea	1	0.41	0.80
	superbowl_ne	0	0.52	0.17
blf-full	hockey_usa	1	0.24	0.25
	hockey_can	0	0.43	0.42
	ausopen_alcaraz	1	0.30	0.45
	ausopen_djokovic	0	0.17	0.30
	superbowl_sea	1	0.53	0.56
	superbowl_ne	0	0.32	0.31
futuresim-full	hockey_usa	1	0.33	0.85
	hockey_can	0	0.62	0.00
	ausopen_alcaraz	1	0.38	1.00
	ausopen_djokovic	0	0.29	0.00
	superbowl_sea	1	0.42	1.00
	superbowl_ne	0	0.36	0.00
react	hockey_usa	1	0.23	0.30
	hockey_can	0	0.44	0.10
	ausopen_alcaraz	1	0.31	1.00
	ausopen_djokovic	0	0.11	0.00
	superbowl_sea	1	0.60	1.00
	superbowl_ne	0	0.38	0.00

Table 5: Mean sequential forecast at the last trajectory date and at the probe. Bold marks an exact hit. blf-full moves by more than 0.05 on only two of six questions.

C Per-date series

harness	series	D_1	D_2	D_3	D_4	D_5
analytica-full	stabilisation	0.85	1.25	1.26	1.06	0.93
	Brier SEQ/PAR	.365/.347	.348/.275	.320/.335	.308/.294	.318/.250
blf-full	stabilisation	1.07	1.59	2.09	1.71	1.13
	Brier SEQ/PAR	.378/.379	.366/.338	.329/.336	.283/.283	.240/.274
futuresim-full	stabilisation	0.68	1.07	1.02	1.33	0.48
	Brier SEQ/PAR	.403/.404	.405/.384	.353/.354	.316/.316	.299/.288
react	stabilisation	0.90	0.67	0.58	0.76	0.59
	Brier SEQ/PAR	.377/.349	.365/.344	.338/.325	.283/.303	.265/.232

Table 6: Indexed by trajectory step rather than calendar date, because the three groups resolve on different dates. Stabilisation is $\text{sd}(\text{SEQ})/\text{sd}(\text{PAR})$.